Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/18/2008
PI Name:	Glazier, James Alexander Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity/Subproject B		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	47405-7105	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	03/01/2003	**End Date:**	02/28/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1619		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this project is to compare fundamental embryological morphogenetic phenomena under normal and zero gravity conditions. Specific areas to be considered include gastrulation, limb development and cell sorting. Tasks include development of biomechanical models for devlopment and experiments under normal and simulated microgravity conditions. These experiments and simulations will be coordinated with supporting and parallel research by Dr. Rusty Lansford (CalTech) and Dr. Gabor Forgacs (U. Missouri, Columbia). Task Objective: Develop protocols necessary for assays of effects of simulated microgravity on cell motility and signaling during developmental processes. Task Significance: Since many simulated microgravity effects on development are quite subtle, it is essential to have a complete range of quantitative techniques to assess changes in cell behavior related to cytoskeletal disruption under		

	microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have discovered a fundamentally new patterning mechanism operating in chick limb beyond classical positional coding and reaction-diffusion. This mechanism has fundamentaly different character and is probably important in a large variety of other developmental processes in other animals. Understanding the mechanisms of development is a key step in developing therapies to treat developmental abnormalities, design regenerative medical therapies and treat degenerative diseases in adults. The microfluidics cell, when further developed will assist in automating and streamlining drug discover and evaluation and may have applications in detecting biotoxins.
Task Progress:	Cell adhesion molecules are widely recognized as playing key roles in development, wound healing and cancer metastasis. However, significant uncertainty remains concerning the exact mechanisms by which differences in the number and types of adhesion molecules give rise to biological structures. We hypothesized that variations in the number of cell adhesion molecules of the same type suffice to induce sorting of cells into coherent clusters, in agreement with the Differentiation Adhesion Hypothesis and previous experiments. Simulated microgravity can induce disorganization of both E-cadherin adherens junctions and cortical actin, which both affect the adhesivity of cadherins. We thus hypothesized that microgravity could affect cell sorting and pattering induced by variations in expression of a single cadherin. If so, microgravity might interfere with cell sorting and morphological patterning in general and thus have serious biomedical consequences during extended space flights. Most of the experimental component of this grant was conducted by co-PIs Dr. R. Lansford at the California Institute of Technology and Dr. G. Forgacs at the University of Missouri, Columbia. Our focus has been on developing computational tools (CompuCell3D) to allow simulation of cell behaviors in tissues and aggregates. We also developed a set of cell lines expressing different levels of an E-cadherin-GFP fusion protein, and investigated single-cadherin cell sorting experimentally at the molecular and tissue levels to confirm that variations of expression of a single cadherin can induce cell sorting under normal gravity. Unfortunately, funding for this project terminated before we were able to conduct the 0g analog experiments. Key Outcomes: 1) Experimental and computational demonstration that continuous gradations of E-cadherin can induce cell sorting in aggregates. 2) Experimental determination of the chondrogenic aggregation of cells in micromass culture under various conditions. 3) Development of open-source software for modeling the development of tissues, CompuCell3D which can be used to investigate the tissue-level effects of microgravity-induced changes in cell adhesion and motility.
Bibliography Type:	Description: (Last Updated: 12/29/2020)
Articles in Peer-reviewed Journals	Glazier JA, Zhang Y, Swat M, Zaitlen B, Schnell S. "Coordinated action of N-CAM, N-cadherin, EphA4, and ephrinB2 translates genetic prepatterns into structure during somitogenesis in chick." Curr Top Dev Biol. 2008;81:205-47. PMID: 18023729 , Jan-2008
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Articles in Peer-reviewed Journals	Merks RM, Brodsky SV, Goligorksy MS, Newman SA, Glazier JA. "Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling." Dev Biol. 2006 Jan 1;289(1):44-54. PMID: 16325173 , Jan-2006
Articles in Peer-reviewed Journals	Merks RMH, Glazier JA. "Dynamic mechanisms of blood vessel growth." Nonlinearity 2006 Jan;19(1):C1-10. http://dx.doi.org/10.1088/0951-7715/19/1/000 , Jan-2006
Articles in Peer-reviewed Journals	Cui C, Yang X, Chuai M, Glazier JA, Weijer CJ. "Analysis of tissue flow patterns during primitive streak formation in the chick embryo." Dev Biol. 2005 Aug 1;284(1):37-47. PMID: 15950214 , Aug-2005
Articles in Peer-reviewed Journals	Sanyal S, Glazier JA. "Viscous instabilities in flowing foams: a Cellular Potts Model approach." J Stat Mech. 2006 Oct;(10):p10008. https://doi.org/10.1088/1742-5468/2006/10/P10008 ; PMID: 19002269; PMCID: PMC2581818 , Oct-2006
Journal/Magazine covers	Merks RMH, Glazier JA. "Cover illustration in journal Nonlinearity for article, “Dynamic mechanisms of blood vessel growth." Nonlinearity 2006 Jan;19(1):C1-10. http://dx.doi.org/10.1088/0951-7715/19/1/000 , Jan-2006